

Width vectors, M -convexity, and the horizontal-strip defect

A refutation, and the theorem that replaces it

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Abstract

Condition (A) — log-concavity in a of the cylindric Kostka number $k(a, b)$ at fixed b — reduces to the statement that a sum of concentric, symmetric, PF_2 sequences Trap_ν , indexed by a slice Σ_b , is PF_2 . The node **A-shape-blind-impossible** proves that any such proof must consume the full *width vectors* $w(\nu)$ and not only the half-widths they determine. The candidate hypothesis for the present session was that the width-vector set $W = \{w(\nu) : \nu \in \Sigma_b\}$ is an M -convex subset of $\mathbb{Z}^m$.

We refute it, for a structural reason, and we locate the reason exactly. On any slice $\sum_i w_i(\nu) = G + m - \|y(\nu)\|_1$ and $\|y\|_1 \equiv \sigma \pmod{2}$, so every width vector of a slice has coordinate sum of one fixed parity (Lemma 7). An M -convex set has constant coordinate sum (Lemma 5) and an $M^\natural$ -convex set has coordinate sums forming an integer interval (Lemma 6); the half-widths occurring on a slice are consecutive integers, so two distinct half-widths put two coordinate sums at distance exactly 2 — a gap no such set can have. Hence M -convexity of W *forces* a single half-width (Theorem 8; the converse holds on every case measured but is not proved, see Remark 9). Measured: of 35 973 nontrivial slices W is M -convex on 13 522 and fails on 22 451, and the failures coincide *exactly* with the multiple-half-width slices. The $M^\natural$ relaxation, the graded lift $\{(k(\nu), w(\nu))\}$ and the level-wise version all fail as well, the first two on the identical 22 451 slices.

The diagnosis is sharper than the refutation. The strata the hypothesis uses are the *level* sets of the horizontal-strip defect $k(\nu) = \sum_i (-y_i)_+$, which is a convex function of ν ; level sets of a convex function are shells, and shells have holes. Its *sublevel* sets are the convex objects, and they live in y -space, where the coordinate sum is constant and M -convexity is well posed. We prove:

Theorem M. *For every box $B = \prod_i [P_i, Q_i] \subset \mathbb{Z}^m$, every $\sigma \in \mathbb{Z}$ and every j , the set $\{y \in B : \sum_i y_i = \sigma, \sum_i (-y_i)_+ \leq j\}$ is M -convex.*

In particular, by **A-M-profile-box-slice**, the horizontal-strip-defect sublevel sets $\Sigma_b^{\leq j}$ of every slice, at every m , are M -convex. The theorem is sharp in a way that matters: the same statement for a *general* separable convex ϕ is false (1308/12081 failures, witness recorded), so it is not a corollary of separable-convex theory — it uses that $t \mapsto (-t)_+$ is 1-Lipschitz and monotone. Tested in the abstract class its hypotheses carve out (all boxes in $[-3, 3]$, all σ , all j , $m = 2, 3, 4$): 109 564 tests, 0 failures.

1 What was asked, and what happened

The brief for this session named one hypothesis and one twenty-minute test. The hypothesis:

Hypothesis 1 ((H1), as briefed). $W = \{w(\nu) : \nu \in \Sigma_b\} \subset \mathbb{Z}^m$ satisfies the symmetric exchange axiom, i.e. is an M -convex set.

Hypothesis 2 ((H2), as briefed). A finite family of concentric interval-convolutions whose width-vector set is M -convex has PF₂ sum.

The test: take the non-PF₂ certificate of **A-shape-blind-impossible** and ask whether *its* width-vector set is M -convex; if yes, (H2) is dead on arrival.

The test was run first, and its answer is recorded in §3: the non-PF₂ certificate is *not* M -convex, so (H2) survives — but the PF₂ certificate is not M -convex either, so the hypothesis excludes *both* sides of the certificate and the test discriminated nothing. It is a non-refutation, not evidence.

(H1), however, is false, and the rest of this note is about why, and about the theorem that stands where it fell.

2 Setup

Throughout $m \geq 1$, $n \geq m$, $G := n - m$, and $\mu \leq \lambda$ are cylindric shapes in $C^{n,m}$: bi-infinite strictly increasing sequences with $x_{i+m} = x_i + n$, stored as $(x_1, \dots, x_m)$. Indices are cyclic with $\lambda_0 = \lambda_m - n$. Write $d = \lfloor \lambda/\mu \rfloor$. We use, as premises, the following results from `proofs/registry/cylindric-lorentzian.json`, with their stored trust levels:

- **A-slice-sum-reformulation** (*proved*): with $L_i(\nu) = \max(\nu_i, \lambda_{i-1} + 1)$, $R_i(\nu) = \min(\nu_{i+1} - 1, \lambda_i)$, $w_i(\nu) = R_i - L_i + 1$ and $\text{Trap}_\nu = *_i \mathbf{1}_{[0, w_i - 1]}$ shifted by $\sum_i (L_i - \nu_i)$,

$$k(a, b) = \sum_{\nu \in \Sigma_b} \text{Trap}_\nu(a), \quad \Sigma_b = \{\nu \in \text{Box}(\mu) : S_\nu = |\mu| + b\}.$$

- **A-general-m-concentricity** (*proved*): $\sum_i (L_i + R_i) = S_\nu + |\lambda|$, so every Trap_ν on a slice is symmetric about the single centre $c = (d - b)/2$.
- **pf2-convolution** (*proved*; a *paper* proof via Toeplitz matrices and Cauchy–Binet, *not* formalised): PF₂ — log-concave with interval support — is closed under convolution. Hence each Trap_ν is PF₂ at every m .
- **A-half-width-is-l1-distance** (*proved*): with $a_i = \lambda_{i-1} + 1$, $y_i = \nu_i - a_i$, $g_i = \lambda_i - \lambda_{i-1} - 1 \geq 0$ (so $\sum_i g_i = G$) and $\sigma = \sum_i y_i$,

$$w_i = g_i + 1 - (y_i)_+ - (-y_{i+1})_+, \quad (1)$$

$$\Lambda(\nu) = \frac{1}{2}(G - \|y\|_1), \quad c - \Lambda(\nu) = k(\nu) := \sum_i (-y_i)_+ \geq 0, \quad (2)$$

and $k(\nu) = 0$ iff λ/ν is a cylindric horizontal strip. Here $(t)_+ = \max(t, 0)$ and Λ is the support half-width of Trap_ν .

- **A-M-profile-box-slice** (*proved*): $\nu \mapsto y$ carries the effective slice Σ_b° (the ν with $\text{Trap}_\nu \neq 0$) bijectively onto

$$Y = \left\{ y \in \prod_i [P_i, Q_i] : \sum_i y_i = \sigma \right\}, \quad P_i = \max(\delta_i, -g_{i-1}), \quad Q_i = g_i - (\delta_{i+1})_-, \quad (3)$$

with $\delta_i = \mu_i - \lambda_{i-1} - 1$ — a *box intersected with a hyperplane*.

- **A-general-m-tent** (*proved*): the half-widths occurring on a slice are *consecutive* integers (gap-free), and the maximum is attained.
- **A-shape-blind-impossible** (*proved*): no condition on the multiset of half-widths alone can imply PF_2 of the sum. Certificate: $\{(1, 1), (2, 2), (1, 5), (2, 6)\}$ and $\{(1, 1), (2, 2), (3, 3), (4, 4)\}$ both have half-widths $0, 1, 2, 3$; the first sums to $(1, 3, 4, 6, 4, 3, 1)$, not PF_2 since $4^2 < 3 \cdot 6$, the second to $(1, 3, 6, 10, 6, 3, 1)$, PF_2 .

Condition (A), the open node **ell3-coupling-gap** in the regime $\ell \geq 3, m \geq 2$, is the assertion that $a \mapsto k(a, b)$ is PF_2 for every b .

Definition 3 (discrete convexity). A finite $W \subset \mathbb{Z}^m$ is M -convex if for all $w, v \in W$ and all i with $w_i > v_i$ there is j with $w_j < v_j$ and $w - e_i + e_j \in W$ and $v + e_i - e_j \in W$. It is $M^\natural$ -convex if for all $w, v \in W$ and all i with $w_i > v_i$, either $w - e_i \in W$ and $v + e_i \in W$, or such a j exists.

3 The first move: the certificate test

Proposition 4. *Neither certificate of A-shape-blind-impossible has an M -convex or $M^\natural$ -convex width-vector set. For the non- PF_2 family $\{(1, 1), (2, 2), (1, 5), (2, 6)\}$ the exchange fails at $(w, v, i) = ((2, 6), (1, 1), 1)$; for the PF_2 family $\{(1, 1), (2, 2), (3, 3), (4, 4)\}$ it fails at $((4, 4), (1, 1), 1)$.*

Proof. Both sets have coordinate sums $\{2, 4, 6, 8\}$. In each case take w the top element and $v = (1, 1)$ and $i = 1$. For the $M^\natural$ axiom, $w - e_1 \notin W$ in both cases, and the only index j with $w_j < v_j$ would need $w_2 < 1$, impossible. The M -convex axiom is stronger, so it fails too. $\square$

So (H2) is *not* refuted by the test the brief designed. But the hypothesis of (H2) is satisfied by *neither* side of the certificate, so the test could not have discriminated: it is a vacuous pass, of the kind that a level count alone would have reported as a success. This is recorded here rather than in a footnote because the whole value of the test was supposed to be its decisiveness.

4 (H1) is false: the parity obstruction

Two lemmas about Definition 3, proved from the local axioms (we do not need the lifting characterisation of $M^\natural$ -convexity, although §9 verifies it).

Lemma 5. *If $W \subset \mathbb{Z}^m$ is finite and M -convex then $\sum_i w_i$ is constant on W .*

Proof. Suppose not. Call a pair $(w, v) \in W \times W$ *admissible* if $\sum_i w_i > \sum_i v_i$; by assumption one exists. Put $\Psi(w, v) = \sum_i (v_i - w_i)_+$ and choose an admissible pair minimising Ψ . Since $\sum w > \sum v$ there is i with $w_i > v_i$.

If $\Psi(w, v) = 0$ then $w_j \geq v_j$ for every j , so no j with $w_j < v_j$ exists and the exchange axiom fails at (w, v, i) — a contradiction. Hence $\Psi(w, v) \geq 1$ and the axiom supplies j with $w_j < v_j$ and $v' := v + e_i - e_j \in W$. Now $\sum v' = \sum v < \sum w$, so (w, v') is admissible, and Ψ changes only in the coordinates i and j :

$$(v_i + 1 - w_i)_+ = 0 = (v_i - w_i)_+ \quad (\text{since } w_i \geq v_i + 1),$$

$$(v_j - 1 - w_j)_+ = v_j - w_j - 1 = (v_j - w_j)_+ - 1 \quad (\text{since } v_j \geq w_j + 1).$$

Hence $\Psi(w, v') = \Psi(w, v) - 1$, contradicting minimality. $\square$

Lemma 6. *If $W \subset \mathbb{Z}^m$ is finite and $M^\natural$ -convex then $\{\sum_i w_i : w \in W\}$ is an integer interval.*

Proof. It suffices to show: if $w, v \in W$ with $\sum w \geq \sum v + 2$ then some $u \in W$ has $\sum u = \sum v + 1$. Call such a pair admissible and choose one minimising $\Psi(w, v) = \sum_i (v_i - w_i)_+$; take i with $w_i > v_i$. The $M^\natural$ axiom gives two alternatives. In the first, $v + e_i \in W$ and $\sum(v + e_i) = \sum v + 1$, which is the conclusion. In the second we get j with $w_j < v_j$ and $v' := v + e_i - e_j \in W$; exactly as in Lemma 5, $\sum v' = \sum v$ so (w, v') is admissible and $\Psi(w, v') = \Psi(w, v) - 1$, contradicting minimality. So the first alternative holds. $\square$

Lemma 7 (parity). *Fix a slice. For every $\nu \in \Sigma_b^\circ$,*

$$\sum_{i=1}^m w_i(\nu) = G + m - \|y(\nu)\|_1 \quad \text{and} \quad \sum_{i=1}^m w_i(\nu) \equiv G + m + \sigma \pmod{2}.$$

In particular all width vectors of one slice have coordinate sum of the same parity, and $\sum_i w_i = 2\Lambda(\nu) + m$.

Proof. Summing (1) over i and using $\sum_i g_i = G$ together with $(t)_+ + (-t)_+ = |t|$ (the index shift $i \mapsto i + 1$ is a cyclic bijection) gives $\sum_i w_i = G + m - \sum_i |y_i|$, which is $2\Lambda + m$ by (2). For the parity, $|y_i| \equiv y_i \pmod{2}$ for every integer y_i , so $\|y\|_1 \equiv \sum_i y_i = \sigma \pmod{2}$, and σ is constant on the slice because S_ν and λ are. $\square$

Theorem 8. *Fix a slice with $\Sigma_b^\circ \neq \emptyset$ and let $W = \{w(\nu) : \nu \in \Sigma_b^\circ\}$. If W is M -convex, or merely $M^\natural$ -convex, then Λ is constant on Σ_b° — the slice carries a single half-width.*

Consequently (H1) fails on every slice carrying two or more distinct half-widths, and it fails for a reason that no amount of information about the width vectors can repair: the set lies in a parity coset.

Proof. Suppose W is M -convex. By Lemma 5 the coordinate sum is constant on W , and by Lemma 7 it equals $2\Lambda + m$, so Λ is constant.

Suppose only that W is $M^\natural$ -convex (a weaker hypothesis, by Definition 3). By Lemma 6 the coordinate sums form an integer interval. By **A-general-m-tent** the occurring half-widths are consecutive integers, so if two occur then $2\Lambda + m$ takes two values differing by exactly 2 — and by Lemma 7 no value of intermediate parity occurs. An interval containing s and $s + 2$ contains $s + 1$, a contradiction. Hence Λ is constant. $\square$

Remark 9 (the converse is measured, not proved). The converse direction is *not* claimed. On all 13 522 measured single-half-width slices with $|W| \geq 2$ the set W is in fact M -convex, with no exception, so Theorem 8 is empirically an equivalence; but we have no proof. The obvious argument does not work: constancy of Λ makes $\|y\|_1$, hence k , constant, so Σ_b° is a single *level* set of k — and Theorem 13 is about *sublevel* sets, while §5 exhibits level sets of k that are not M -convex. One would additionally need the width map (1) to carry that particular level set to an M -convex image, and (1) is only piecewise affine, affine on each sign-constant region of y . Nothing in this note needs the converse: the refutation uses only the two implications proved above.

Remark 10 (the measurement). Over 77 866 nonempty slices ($m = 2, \dots, 6$; $G = n - m$ from 0 to 9; see §9 for the plan and the object-size census), 35 973 have $|W| \geq 2$ and hence give the exchange axiom something to say. Of those, W is M -convex on 13 522 and fails on 22 451; the number of slices carrying two or more coordinate sums is 22 451 — *exactly* the failures. The $M^\natural$ counts are identical (13 522 / 22 451). So on real slices the parity obstruction of Theorem 8 is not merely *an* obstruction, it is the *only* one.

5 The three repairs, and why each fails

Repair 1: $M^{\natural}$ -convexity

Dead by Theorem 8(ii), on the identical 22 451 slices.

Repair 2: the graded lift

Lemma 7 says the obstruction is that the grading $\sum_i w_i$ moves in steps of 2. The defect $k(\nu)$ of (2) moves in steps of 1, so the lift

$$\widehat{W} = \{(k(\nu), w(\nu)) : \nu \in \Sigma_b^\circ\} \subset \mathbb{Z}^{1+m}, \quad \sum \widehat{w} = G + m - \sigma - k(\nu),$$

has coordinate sums forming an integer interval — verified on all 35 973 nontrivial slices, 0 exceptions, so the parity obstruction really is removed. And yet $\widehat{W}$ is $M^{\natural}$ -convex on exactly 13 522 of them and fails on the same 22 451. Smallest witness: $m = 2$, $n = 4$, $\mu = (0, 2)$, $\lambda = (1, 4)$, $b = 1$, $\widehat{W} = \{(0, 1, 3), (1, 1, 1)\}$, exchange failing at $i = 3$ — $(0, 1, 2) \notin \widehat{W}$ and $(1, 1, 2) \notin \widehat{W}$. Removing the parity obstruction does not remove the holes.

Repair 3: level-wise M -convexity

On a fixed level $\{k(\nu) = k\}$ the coordinate sum $G + m - \sigma - 2k$ is constant, so

$$(H1') \quad W_k = \{w(\nu) : k(\nu) = k\} \text{ is } M\text{-convex for each } k$$

is well posed. It is also false: 830 failures among 28 479 levels of size ≥ 2 . Smallest witness: $m = 2$, $n = 6$, $\mu = (0, 3)$, $\lambda = (3, 6)$, $b = 2$, $k = 1$, where $W_1 = \{(1, 3), (3, 1)\}$ — both of coordinate sum 4, with the midpoint $(2, 2)$ absent. Note that this is a failure at $m = 2$, where condition (A) is *proved* (**A-m2-complete**): so (H1') is not merely unproved, it is false in a regime where the conclusion holds, and therefore cannot be a necessary condition either.

Worse, (H1') is *insufficient* even where it holds. In the non-PF₂ certificate of **A-shape-blind-impossible** the four width vectors have four *distinct* half-widths, so every level is a singleton and (H1') is satisfied while PF₂ fails. Thus:

Proposition 11. *Any hypothesis on W that constrains each half-width level separately — in particular any hypothesis satisfied by all families of singleton levels — is insufficient to imply PF₂ of the sum, in the abstract class of concentric interval-convolution families.*

Remark 12 (which class). Proposition 11 is a statement about the *abstract* class, not about slices, and this must be said plainly. **A-shape-blind-impossible**'s first certificate has width differences 0, 0, −4, −4 and is not realisable as a slice at $m = 2$; and no slice-realisable counterexample can exist, since condition (A) is conjectured true and has survived 4 323 908 slice tests. A hypothesis can therefore be refuted only abstractly, and a proof must use slice-specific information that the abstract class does not carry. That asymmetry is the content of the next section.

6 Theorem M: the sublevel sets are M -convex

The unified diagnosis of §5 is a single sentence. The defect $k(\nu) = \sum_i (-y_i)_+$ of (2) is a *convex* function of y (a sum of convex functions of one coordinate each). The strata used by (H1) and

(H1') are its *level* sets — the shells of a convex function. Shells have holes, and consecutive shells sit at distance 2 in $\sum_i w_i$. Asking a shell to be M -convex is asking a sphere to be a ball.

The *sublevel* sets are the convex objects. And they live in y -space, where $\sum_i y_i = \sigma$ is constant and M -convexity is well posed.

Theorem 13 (Theorem M). *Let $B = \prod_{i=1}^m [P_i, Q_i] \subset \mathbb{Z}^m$ be a box, let $\sigma \in \mathbb{Z}$, and put $k(y) = \sum_i (-y_i)_+$. Then for every $j \in \mathbb{Z}$ the set*

$$S_j = \left\{ y \in B : \sum_i y_i = \sigma, \quad k(y) \leq j \right\}$$

is M -convex.

Proof. Let $y, y' \in S_j$ and let i satisfy $y_i > y'_i$. Put $D = \{l : y_l < y'_l\}$; since $\sum_l y_l = \sum_l y'_l$ and $y_i > y'_i$, $D \neq \emptyset$, and $i \notin D$. For $l \in D$ set

$$y^{(l)} = y - e_i + e_l, \quad y'^{(l)} = y' + e_i - e_l.$$

Box and hyperplane. Both have coordinate sum σ . For membership in B : $y_i - 1 \geq y'_i \geq P_i$; $y_l + 1 \leq y'_l \leq Q_l$; $y'_i + 1 \leq y_i \leq Q_i$; $y'_l - 1 \geq y_l \geq P_l$. So $y^{(l)}, y'^{(l)} \in B$ for *every* $l \in D$, and only the constraint $k \leq j$ can fail. (This already proves the case $j = \infty$: a box intersected with a hyperplane is M -convex.)

The increments. Since $t \mapsto (-t)_+$ changes by 0 or 1 under a unit step,

$$k(y^{(l)}) - k(y) = p + q_l, \quad k(y'^{(l)}) - k(y') = p' + q'_l,$$

where, directly from the definition,

$$p = \mathbf{1}[y_i \leq 0], \quad q_l = -\mathbf{1}[y_l \leq -1], \quad p' = -\mathbf{1}[y'_i \leq -1], \quad q'_l = \mathbf{1}[y'_l \leq 0].$$

Note $p, q'_l \in \{0, 1\}$ and $p', q_l \in \{-1, 0\}$, and that p, p' depend only on i .

Case 1: $y_i \geq 1$ and $y'_i \leq -1$. Then $p = 0$ and $p' = -1$, so $k(y^{(l)}) - k(y) = q_l \leq 0$ and $k(y'^{(l)}) - k(y') = -1 + q'_l \leq 0$. Every $l \in D$ works.

Case 2: $y_i \geq 1$ and $y'_i \geq 0$. Then $p = p' = 0$. The y side is safe for every l , since $k(y^{(l)}) - k(y) = q_l \leq 0$. For the y' side, $k(y'^{(l)}) = k(y') + q'_l$. If $k(y') \leq j - 1$, every $l \in D$ works. If $k(y') = j$ we need $q'_l = 0$, i.e. an $l \in D$ with $y'_l \geq 1$. Suppose there is none, so every $l \in D$ has $y'_l \leq 0$ and hence $y_l \leq y'_l - 1 \leq -1$. Then

$$k(y) - k(y') = \sum_t \left((-y_t)_+ - (-y'_t)_+ \right).$$

For $t \in D$ both $y_t, y'_t \leq 0$, so the term is $y'_t - y_t \geq 1$. For $t \notin D$ we have $y_t \geq y'_t$, so the term is $\geq -(y_t - y'_t)$ because $(-\cdot)_+$ is 1-Lipschitz; and at $t = i$ the term is exactly $0 - 0 = 0$, strictly greater than $-(y_i - y'_i) < 0$. Summing and using $\sum_{t \in D} (y'_t - y_t) = \sum_{t \notin D} (y_t - y'_t)$,

$$k(y) - k(y') > \sum_{t \in D} (y'_t - y_t) - \sum_{t \notin D} (y_t - y'_t) = 0,$$

so $k(y) > k(y') = j$, contradicting $y \in S_j$. Hence such an l exists.

Case 3: $y_i \leq 0$ (hence $y'_i \leq -1$). Then $p = 1$ and $p' = -1$. The y' side is safe for every l , since $k(y'^{(l)}) - k(y') = -1 + q'_l \leq 0$. For the y side, $k(y^{(l)}) = k(y) + 1 + q_l$. If $k(y) \leq j - 1$, every $l \in D$ works. If $k(y) = j$ we need $q_l = -1$, i.e. an $l \in D$ with $y_l \leq -1$. Suppose there is none, so

every $l \in D$ has $y_l \geq 0$ and hence $y'_l > y_l \geq 0$. Then for $t \in D$ both $(-y_t)_+$ and $(-y'_t)_+$ vanish, contributing 0; for $t \notin D$ we have $y_t \geq y'_t$ and so $(-y_t)_+ - (-y'_t)_+ \leq 0$; and at $t = i$, where $y_i \leq 0$ and $y'_i \leq -1$ and $y_i > y'_i$, the term is $-y_i + y'_i < 0$ strictly. Hence $k(y) < k(y') \leq j$, contradicting $k(y) = j$. Such an l exists.

The three cases are exhaustive: either $y_i \leq 0$ (Case 3, and then $y'_i < y_i \leq 0$ gives $y'_i \leq -1$), or $y_i \geq 1$ and then $y'_i \leq -1$ (Case 1) or $y'_i \geq 0$ (Case 2). In each case we produced $l \in D$ with $y^{(l)}, y'^{(l)} \in S_j$, which is the symmetric exchange axiom. $\square$

Remark 14 (Case 3 is Case 2 under a symmetry). The two nontrivial cases are not independent. On the hyperplane $\sum_i y_i = \sigma$ we have $k(-y) = \sum_i (y_i)_+ = k(y) + \sigma$, so $y \mapsto -y$ carries S_j for (B, σ, j) bijectively onto $S_{j+\sigma}$ for $(-B, -\sigma, j + \sigma)$, where $-B = \prod_i [-Q_i, -P_i]$ is again a box. Composing this with the swap of the roles of y and y' — which is legitimate because $y_i > y'_i$ becomes $(-y'_i)_i > (-y)_i$ — sends the hypothesis of Case 2 ($y_i \geq 1, y'_i \geq 0$) to that of Case 3 ($(-y'_i)_i \leq 0$) and back, and fixes Case 1. Since Theorem 13 is quantified over all boxes, all σ and all j , Case 3 follows from Case 2, and the proof really spends one argument, once, plus a symmetry. Verified: the involution maps the sublevel set onto its target in all 17 096 cases tested, and exchanges the case labels on all 206 010 encountered triples (y, y', i) (Case 1 fixed on 35 216, Case 2 $\leftrightarrow$ 3 on 103 005 each way); the two cases' instrumented counts in §9 agree exactly, which is how the symmetry was noticed.

Corollary 15. *For every $m \geq 1$, every cylindric $\mu \leq \lambda$ and every b , and every $j \geq 0$, the horizontal-strip-defect sublevel set*

$$\Sigma_b^{\leq j} = \{\nu \in \Sigma_b^\circ : c - \Lambda(\nu) \leq j\}$$

is carried by $\nu \mapsto y$ onto an M -convex subset of $\mathbb{Z}^m$. In particular $\Sigma_b^{\leq 0}$, the set of ν with λ/ν a cylindric horizontal strip, is M -convex, and so is the whole effective slice.

Proof. Immediate from Theorem 13 and (3): the image of Σ_b° is $B \cap \{\sum y_i = \sigma\}$, and (2) identifies $c - \Lambda$ with k . $\square$

7 Sharpness: Theorem M is not separable-convex theory

The proof of Theorem 13 uses two properties of $\phi_i(t) = (-t)_+$ beyond convexity: it is 1-Lipschitz, which bounds the increments by 1 and powers the Lipschitz comparison in Case 2, and it is non-increasing, which fixes the signs of p, p', q, q' . Both are load-bearing:

Proposition 16. *The analogue of Theorem 13 for a general separable convex $\phi(y) = \sum_i \phi_i(y_i)$ is false. Witness: $m = 3$, $B = [-2, -1] \times [-2, 1] \times [-2, 1]$, $\sigma = -1$, and a separable convex ϕ for which the sublevel set at level 6 is*

$$S = \{(-2, 1, 0), (-1, -1, 1), (-1, 0, 0), (-1, 1, -1)\},$$

where the exchange fails at $((-1, -1, 1), (-2, 1, 0), i = 1)$: the only l with $y_l < y'_l$ is $l = 2$, and $(-1, -1, 1) - e_1 + e_2 = (-2, 0, 1) \notin S$.

Over four independent random separable convex ϕ on $m = 3$ with boxes in $[-2, 2]$, the failure rates were 1308/12081, 798/10334, 1392/12006 and 1116/11879. So Theorem 13 is not an instance of a general statement about separable convex functions on base polyhedra; the specific shape of $(-t)_+$ is used.

Two further hypotheses are not load-bearing, and saying so prevents a false reading. First, $\phi_i(t) = (c_i - t)_+$ with arbitrary integer shifts c_i reduces to Theorem 13 by the translation $z = y - c$, which maps boxes to boxes and shifts σ ; so does $\sum_i |y_i - c_i|$, since on the hyperplane $\sum_i |z_i| = \sigma' + 2 \sum_i (-z_i)_+$. Second, $\max_i |y_i| \leq j$ is a box constraint, so the sublevel sets of $\max_i |y_i|$ are again box-cap-hyperplane and carry no new information. Each of these came back 100% green in testing, and none of them is a control.

8 What the right hypothesis must see

Collecting §4–§6 into a statement about the search space:

1. *No class with gap-free gradings.* Any hypothesis of the form “ W belongs to a class whose members have gap-free coordinate-sum sets” — M -convex, $M^\sharp$ -convex, and anything between — is *never satisfied* by a slice with two half-widths (Theorem 8). Such a hypothesis is not wrong; it is empty. This is the insufficiency failure mode rather than the falsity one: the class cannot close the goal because the objects are not in it.
2. *No level-wise class.* Any hypothesis satisfied by all singleton-level families is insufficient (Proposition 11), because **A-shape-blind-impossible**’s non-PF₂ certificate has one width vector per level. So the condition must couple levels.
3. *No condition on the set W alone can be both.* Items 1 and 2 pincer the hypotheses expressible in terms of the image set $W \subset \mathbb{Z}^m$: the condition must be cross-level, and it must tolerate parity gaps and holes. What survives is a condition on the *parametrisation* $\nu \mapsto w(\nu)$ — on the pair (Y, w) — rather than on its image. Theorem 13 says the domain side of that pair is as good as it could be: Y is M -convex and so is every horizontal-strip-defect sublevel set of it, at every m . A hypothesis stated in y -coordinates also escapes Remark 12: the abstract class of **A-shape-blind-impossible** has no y , so it cannot refute one.

In generating-function form, what remains to be proved is this. With $q = e^{2\theta}$ and $[w]_q = 1 + \dots + q^{w-1}$, concentricity and (2) give

$$q^{-c} \sum_a k(a, b) q^a = \sum_{y \in Y} q^{-\Lambda(y)} \prod_{i=1}^m [w_i(y)]_q = \frac{1}{\sinh^m \theta} \sum_{y \in Y} \prod_{i=1}^m \sinh(w_i(y) \theta),$$

a sum, over the M -convex set Y of (3), of products of m hyperbolic sines whose arguments are the piecewise-affine functions (1). Condition (A) is the log-concavity of the coefficients of the left side. The stratification by k is now the natural one on both sides: k moves in unit steps, $\Lambda = c - k$, and the k -th stratum is supported on $|t| \leq c - k$, so the strata are *nested* — and Corollary 15 says each nest $\Sigma_b^{\leq j}$ is M -convex. We do not close (A) here and we do not claim progress on it beyond this reorientation.

9 Computational verification

All code is in `proofs/code-1004-wvec/`. Every instrument was calibrated against a value fixed in advance before being used on an unknown, and every absence claim carries a positive control.

Instrument calibration. `mconv.py` implements Definition 3 and was run first on seven sets whose status was written down in advance: the bases of $U_{2,3}$ (M -convex), $\{(2, 0, 0), (0, 2, 0)\}$ (neither), a box (not M -convex, $M^\natural$ -convex), box-cap-hyperplane (M -convex), $\{(0, 0), (1, 1)\}$ (neither), a simplex (not M -convex, $M^\natural$ -convex), and a singleton (both). All seven agreed. `firstmove.py` reproduced both recorded values of **A-shape-blind-impossible** — the sums $(1, 3, 4, 6, 4, 3, 1)$ and $(1, 3, 6, 10, 6, 3, 1)$, and their PF_2 status — from the width vectors, before the M -convexity test was run on them.

Lemma 7. The identity $\sum_i w_i = G + m - \|y\|_1$ holds on all 141 759 nonzero summands tested, and the parity is constant on each slice and equal to the predicted $(G + m + \sigma) \bmod 2$ on all 77 866 slices. 0 failures.

Theorem 8 and the repairs. Sweep plan (`sweep.py`): $m = 2$ ($n \leq 11, d \leq 12$), $m = 3$ ($n \leq 10, d \leq 10$), $m = 4$ ($n \leq 9, d \leq 9$), $m = 5$ ($n \leq 8, d \leq 8$), $m = 6$ ($n \leq 8, d \leq 7$) — 77 866 nonempty slices, with $m = 2$: 11 452, $m = 3$: 34 977, $m = 4$: 27 350, $m = 5$: 3 486, $m = 6$: 601, and $G = n - m$ ranging over $0, \dots, 9$ with 11 461 slices at $G = 6$, 18 856 at $G = 7$, 2 753 at $G = 8$ and 3 476 at $G = 9$. Object-size census, reported because a count alone hides vacuity: $|W| = 1$ on 41 893 slices (where every exchange axiom is vacuously true and which are therefore excluded from all M -convexity counts), $|W| = 2$ on 18 958, 3 on 10 751, 4 on 3 335, 5 on 1 773, 6 on 742, 7 on 306, ≥ 8 on 108; number of levels 1 on 55 415, 2 on 18 970, 3 on 3 255, 4 on 222, 5 on 4; largest level of size 1 on 52 564, ≥ 2 on 25 302. Results are in Remark 10 and §5.

Lemmas 5 and 6. Exhaustive over all subsets of $[0, 3]^2$, $\{0, 1\}^3$, $\{0, 1\}^4$ and $[0, 2]^2$. Lemma 5: 114 M -convex subsets found, none with non-constant coordinate sum. Lemma 6: among 27 923 + 29 + 4 930 subsets with a *gap* in their coordinate sums, 0 are $M^\natural$ -convex, while the positive control found 230, 51 and 490 $M^\natural$ -convex subsets with ≥ 2 distinct sums — so the search was not vacuous. Independently, the lifting characterisation (W is $M^\natural$ -convex iff $\{(-\sum w, w)\}$ is M -convex) agreed with Definition 3 on all 131 836 subsets tested, 0 disagreements; the proofs above do not use it.

Theorem 13, in the abstract class. This is the test that matters, because a sufficient condition that is empirically perfect on instances can still be false in the class its own hypotheses carve out. Over *all* boxes $\prod_i [P_i, Q_i]$ with $P_i \leq Q_i$ in $[-3, 3]$ (resp. $[-2, 2]$), *all* σ and *all* j : $m = 2, [-3, 3]$: 691 tests; $m = 3, [-2, 2]$: 4 994; $m = 3, [-3, 3]$: 89 235; $m = 4, [-2, 2]$: 14 644. Total 109 564 nontrivial sublevel sets, 0 failures, largest $|S|$ tested 67. Positive control: the base sets $B \cap \{\sum = \sigma\}$ came back M -convex in all 132 477 cases. On real slices (`controls.py`): 15 058/15 058.

Refusal panel for Theorem 13. Every outcome below was predicted before the run.

- Deleting one point from Y must break M -convexity: broke on 4 155 of 25 389 (the instrument can say no; the rate is low because most Y are small).
- *Superlevel* sets $\{k \geq j\}$ must break: broke on 1 756 of 13 216. This is the control that gives Theorem 13 content — sublevel 15 058/15 058 against superlevel 11 460/13 216.
- A non-convex stratification $\phi(y) = \#\{i : y_i < 0\}$ must break: broke on 1 347 of 12 256; witness $\{(-2, 0), (0, -2)\}$ with $(-1, -1)$ absent.
- A general separable convex ϕ must break if Theorem 13 is sharp: Proposition 16.

A control that was no control. The first refusal panel for Theorem 13 stratified by $\sum_i (-y_i)_+$, by $\sum_i (y_i)_+$ and by $\|y\|_1$, and all three returned 11 580/11 580 — *identical* counts. They are the same predicate: on a slice $\sum_i y_i = \sigma$ is constant, so $\sum_i (-y_i)_+ = \frac{1}{2}(\|y\|_1 - \sigma)$ and $\sum_i (y_i)_+ = \frac{1}{2}(\|y\|_1 + \sigma)$ are affine in one another and induce the same ordered stratification of Y — verified, 3 099/3 099 slices. The panel was one test run three times. It was replaced by the panel above. This is recorded because the three green columns would otherwise read as triple confirmation.

Verifying the proof, not the statement. `proofcheck.py` re-runs the three cases of Theorem 13’s proof over the abstract class, and for every encountered (S_j, y, y', i) records which case fires, checks that the candidate set the proof *names* is nonempty (the existence claims of Cases 2 and 3), and checks that *every* named candidate really lands in S_j . All three cases fire. The first run reported 3 336 972 failures, all in Case 2; the cause was a transcription error in the instrument and not in the proof — the slack shortcut in Case 2 belongs to the y' side ($k(y') < j$) and had been coded on the y side ($k(y) < j$). Case 3, whose slack genuinely is on the y side, reported 0 failures on the same run. Corrected results: 0 failures over 6 257 408 Case-1, 14 914 557 Case-2 and 14 914 557 Case-3 triples, with every named candidate landing in S_j ($10\,810\,814 + 19\,033\,146 + 19\,033\,146$ exchange checks) and no existence claim failing. The exact agreement of the Case-2 and Case-3 columns is what revealed Remark 14.

10 Gaps and status

1. **Condition (A) is not closed, at any $m \geq 2$ beyond the proved $m = 2$.** Nothing in this note bears on the open regime of `ell3-coupling-gap` except by removing a candidate route and reorienting the stratification. Theorem 13 is a statement about the domain Y ; (H2) — or any replacement — is still unproved, and we have given no sufficient condition for PF_2 of the sum.
2. **(H2) is neither proved nor refuted.** The brief’s test was a vacuous pass (Proposition 4). Worse, Theorem 8 shows (H2) is *unusable as stated*: its hypothesis is never satisfied by a multi-half-width slice. Grading (H2) as a live conjecture would be an error.
3. **Proposition 11 is abstract-class only** (Remark 12). No slice-realisable counterexample to condition (A) exists or can exist while (A) is true, so the pincer of §8 constrains the *hypotheses* we may write, not the slices.
4. **The converse of Theorem 8 is measured, not proved** (Remark 9): 13 522/13 522 single-half-width slices with $|W| \geq 2$ have W M -convex, with no proof, and the obvious argument fails for a stated reason. Nothing in this note rests on it.
5. **Theorem 13 is proved here on paper and is not formalised.** It rests on nothing but Definition 3 and the identity (3), which is `A-M-profile-box-slice`, *proved*.
6. **pf2-convolution is cited as *proved (paper proof)*, not `lean-verified`.** It is used only to say that each Trap_ν is PF_2 , which is background here, not load-bearing for any theorem of this note.
7. **Open, and the natural next question.** Does M -convexity of the nest $\{\Sigma_b^{\leq j}\}_{j \geq 0}$, together with the explicit width map (1), imply PF_2 of $\sum_{y \in Y} \prod_i [w_i(y)]_q$? Theorem 13 supplies the

hypothesis; nothing here supplies the implication, and §8 item 2 warns that any form of it must couple the strata rather than treat them one at a time.